Hong Ong

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Professional Summary

Software Engineering graduate (First Class Honours, UTS) with nearly a year of production experience. Sole software engineer on OnlyFacts' Ruby on Rails/PostgreSQL climate data platform, and developer on The a2 Milk Company's multi-market Astro/TypeScript websites. Full-stack across data pipelines, APIs, CMS-driven components, localisation and automated testing, with hands-on experience maintaining CI/CD pipelines.

Work Experience

The a2 Milk Company Sydney, NSW
Software Developer (Contract) *June 2026 – Present*

- Developed features for multi-market Astro 5 SSR sites (AU, NZ, SG, HK, VN, CA, US, China) with TypeScript, Kontent.ai (headless CMS), Alpine.js, Algolia and Tailwind CSS on AWS
- Shipped 125+ commits across 3 production repositories, delivering 12+ reusable CMS-driven components with documentation and Vitest unit and smoke tests
- Built a modular article system of 7 CMS-driven components for a health-professional research hub, launching 6 articles to production
- Localised sites for English, Vietnamese and Chinese audiences with a tested UI-label translation utility, and integrated Osano cookie consent and GTM

OnlyFacts Sydney, NSW
Full-Stack Engineer (Contract) *January 2026 – Present*

- Managed all engineering on the Ruby on Rails/PostgreSQL climate data platform as sole software engineer since mid-2026, working with a 4–6 person data team
- Replaced Highcharts maps with a custom D3.js engine (council-area and world maps, hexbin view, zoom, legend) and refactored 1,300+ lines of legacy code
- Developed data pipelines: web scrapers (vehicle sales, carbon credits) and recurring import jobs for emissions and energy datasets
- Shipped a Cloudflare Worker contact API (validation, Google Sheets logging, email alerts, CORS allowlist) and GA4 event tracking for the live React/Vite OnlyFacts Studio site

Software Engineering Intern *November 2025 – January 2026*

- Built and presented 3 feature prototypes to stakeholders, which informed later architecture decisions
- Identified technical debt and maintainability issues in the legacy codebase and presented improvement recommendations adopted in later development

Optik Consultancy (Philips Dynalite) Sydney, NSW
Software Engineering Consultant *May 2024 – February 2025*

- Led a 4–5 person team of software and mechanical engineers to deliver client automation solutions on time and within budget
- Programmed OMRON TM5 cobot using TM FLOW to automate relay/terminal placement, cutting production time by 40% and error rates by 60%

Education

University of Technology Sydney Sydney, NSW
Bachelor of Engineering (Honours), Software Engineering *February 2022 – December 2025*

- First Class Honours | WAM: 87.10 (High Distinction Average) | GPA: 6.59/7.0 | Dean's List (3 consecutive years) | Undergraduate Academic Excellence International Scholarship
- Honours Capstone (HD, showcase nominee): built a C++/Python AST-based plagiarism detector that flagged 8 confirmed misconduct cases over 406 submissions, 73% faster with 73% less storage than text matching
- Relevant Coursework: Database Fundamentals (100), Cloud Computing & SaaS (97), Software Innovation Studio (96), Software Analysis Studio (94), Software Architecture (93), Systems Testing and Quality Management (89)

Projects

VR Career Exploration: Lifeguard Simulation August 2025 – December 2025
Lead Programmer & VR Developer

- Built immersive VR training simulation in Unity/C# using XR Interaction Toolkit; implemented a VR controller-based selection system with haptic, audio, and visual feedback cues, achieving 95% selection accuracy among first-time users during a live showcase
- Developed a dynamic NPC system with randomised spawning; optimised performance for VR headsets using object pooling and reduced draw calls during NPC interactions, cutting frame drops by roughly 50% measured via Unity profiler
- Led a 7-person Agile team, conducting code reviews and technical QA across 3D modelling, animation, and UI work

Thrive X Mobile App Prototype February 2025 – April 2025
Full-Stack Developer

- Built a cross-platform mobile app in Flutter (Dart) integrating Google Gemini API to power AI-driven skill assessments and personalised feedback, spanning front-end, REST APIs, Firebase, and AI workflow orchestration
- Implemented real-time data sync via Firebase (30% faster load times), RESTful API integration, and user authentication/profile management

Leadership Activities

UTS Malaysian and Singaporean Students' Association (MASSA) Sydney, NSW
Treasurer *November 2022 – June 2024*

- Managed $40,000+ budget across 20+ events, consistently achieving break-even or profit targets through strategic financial planning and resource allocation

Technical Skills

Programming Languages: JavaScript, TypeScript, Ruby, Python, SQL, Java, C, C#, C++

Frameworks: Ruby on Rails, ReactJS, Node.js, Astro, Alpine.js, Flutter, D3.js, Tailwind CSS

CMS & Search: Kontent.ai (headless CMS), Algolia

APIs & Architecture: RESTful API design & integration, MVC, object-oriented programming (OOP), background job processing (Heroku Scheduler)

Databases: PostgreSQL, MySQL, Firebase Realtime Database

Cloud: AWS (CloudFront, Elastic Beanstalk, CodeBuild, S3), Google Cloud Platform (GCP)

DevOps: Git, GitHub, Docker, CI/CD pipelines (GitHub Actions, AWS CodeBuild), Agile/Scrum, VS Code, JetBrains Suite

Testing: Unit & integration testing, test-driven development (TDD); pytest, Google Test (C++) from coursework and personal projects

AI & Developer Tooling: Google Gemini, Claude (Claude Code), AI workflow integration, GitHub Actions automation

Languages

Spoken Languages: English, French (Fluent); Vietnamese, Mandarin, Cantonese (Basic)